A-B. Connect properties and constraints

Solutions page 1 | Check after attempting. All hand tables and assessment cases are synthetic teaching data.

A. Retrieval and assumptions

A1. x_1: liquid mole fraction; y_1: vapor mole fraction; z_1: overall mole fraction. γ_1 is the liquid activity coefficient on the chosen pure-liquid standard state; activity $a_1=\gamma_1 x_1$.

A2. $\gamma_1=(0.704857144627\times158.707198211)/(0.50\times150)=$**1.49154537**. $\exp(0.399813)=1.49154575$; the small difference is rounding. The inference assumes ideal vapor, negligible Poynting correction and valid saturation-pressure correlations.

A3. A small residual establishes reproduction by the selected model under its property assumptions. It does not establish thermodynamic consistency at all compositions or predictive accuracy on independent observations.

B1. The product-rule bridge

$g=x \ln \gamma_1+(1-x)\ln \gamma_2$

$dg/dx = \ln \gamma_1-\ln \gamma_2 + x\, d \ln \gamma_1/dx + (1-x)d \ln \gamma_2/dx$

At fixed T and P, Gibbs-Duhem makes the last two terms sum to zero. Thus **$dg/dx=L=\ln(\gamma_1/\gamma_2)$**. Both coefficient functions are constrained together; a single isolated γ value cannot test this composition-dependent identity.

B2-B3. The path determines what can be assessed

Situation	Consequence
Fixed T and P	dg/dx=L; a finite-difference residual is approximate.
Isothermal VLE with varying P	Teaching diagnostic neglects $v^E/(RT)\,dP$. The temperature term is zero.
Real isobaric VLE without independent h^E	No corrected local verdict without independent h^E. Record not assessed.

General path: $\Sigma x_i\, d \ln \gamma_i = -h^E\, dT/(RT^2) + v^E\, dP/(RT)$. Here h^E and v^E are molar excess properties. Isothermal does not automatically mean isobaric.

Not applicable: method does not match the path. **Not assessed:** information/correction is missing or disabled. **Outside this criterion:** an applicable evaluated diagnostic is outside its stated threshold. The first two are neither successful tests nor failures of the observations.

C1-C3. Signed area, not normalized D

Interval	Baseline trapezoid	After uniform +0.20
0-0.25	0.375	0.425
0.25-0.50	0.125	0.175
0.50-0.75	-0.125	-0.075
0.75-1.00	-0.375	-0.325
Signed sum S	0	0.200

Baseline positive area=0.5; signed negative area=-0.5. Endpoint limits were supplied. $D=100|A-B|/(A+B)$, with nonnegative area magnitudes A/B, differs from signed S and is undefined when A+B vanishes. Opposing local errors can cancel.

C-D. A numerical residual is evidence

Solutions page 2 | Use the g-first operator throughout: form g, difference g, then subtract L at the center.

C4. Worked table at x=0.50

Clean g values: 0.40(0.72)+0.60(0.32)=**0.48**; g(0.50)=**0.50**; g(0.60)=**0.48**. L(0.50)=0.50-0.50=**0**. Central slope=(0.48-0.48)/0.20=**0**, so r=**0**.

Changed upper value: g(0.60)=0.60(0.36)+0.40(0.72)=**0.504**. The center L remains zero. Slope=(0.504-0.480)/0.20=**0.120**; r=**+0.120**. Use the composition weight 0.60 when propagating the changed log coefficient.

C5. Independent table at x=0.40

Clean g values are **0.315, 0.360, 0.375**. Slope=(0.375-0.315)/0.20=**0.300**; center L=0.540-0.240=**0.300**, so r=**0** apart from floating-point roundoff.

After changing upper ln γ_2 to 0.425, g(0.50)=0.50(0.375)+0.50(0.425)=**0.400**. Slope=(0.400-0.315)/0.20=**0.425**; r=0.425-0.300=**+0.125**. The center values have not changed.

C6. These uniform-grid derivatives are approximations. Differencing magnifies noise and depends on spacing and rounding; narrower spacing need not help noisy measurements. One local r is not 100 mean |r| across all interior points, and a three-point table supplies no full-range area verdict. Do not apply a dataset-average threshold to a single hand residual.

D1-D4. Actual Lab 03 reference values

Synthetic isothermal run	D (%)	Local score	Result
Clean / 41	0.078692	0.117763	Both meet
Canceling / 41 / 0.25	0.073552	15.598698	Area meets; local outside
Localized / 41 / 0.25	2.013463	1.671839	Both meet; peak remains
Localized / 161 / 0.25	1.941724	1.565607	Both meet; inspect peak

Criteria: D<5%; 100 mean |r|<5. The local score is not a probability or percentage of faulty observations. Clean numerical scores are small rather than identically zero because integration/differentiation is discrete.

D2: the perturbation f=0.25 cos(2πx) has zero full-range integral. Adding (1-x)f to ln γ_1 and -xf to ln γ_2 leaves their weighted g unchanged but changes L by f. The continuous residual changes by -f. **D3-D4:** a narrow defect remains visible although both aggregate criteria are met. Sampling alters numerical representation; it adds no experimental measurements.

D5: at clean / 41, $\Delta p \approx 1.12\times10^{-6}$% and $\Delta y \approx 6.48\times10^{-7}$ percentage points. Near-zero reproduction is expected because NRTL is also the generator family. Herington is not applicable to this isothermal path. A failed fit supplies no modeling verdict.

Optional D5 reopen: use **Study workspace · reopen and compare cases**, load the exported JSON, then select **Restore** beside the case. Check its inputs and rerun **Run modeling test →**. The imported comparison preserves recorded evidence; it does not replace a current fit.

E-G. Evidence for a stable split

Solutions page 3 | Distinguish a homogeneous-feed classification from the final equilibrium state.

E1-E2. An acceptable record identifies axes, case, amplitude, resolution, a visible oscillation/peak, both scores and a limitation. Example: “At 350 K, 41 points and canceling amplitude 0.25, D=0.07355% meets the area criterion while the local score 15.5987 is outside 5. The oscillating residual shows what the integral conceals. This inference assumes ideal vapor and neglected pressure corrections; derivatives also depend on sampling.”

F1-F3. Global support and the hand lever rule

Equal chemical potentials or equal slopes identify a candidate. Stable equilibrium minimizes total Gibbs energy among admissible states at fixed T,P and material balance. Distinct endpoints and a common tangent supporting the whole admitted curve are required. Positive local curvature cannot alone establish the global minimum.

For supplied endpoints 0.20/0.80: z=0.50 gives fβ=fα=0.50; z=0.35 gives fβ=0.25,fα=0.75. Balances: 0.50(0.20)+0.50(0.80)=0.50 and 0.75(0.20)+0.25(0.80)=0.35. These are not Lab 04 endpoints.

G1-G2. Three feeds, one-parameter Margules A=3

z_1	Homogeneous feed / curvature	Equilibrium	Energy decrease / RT
0.02	Stable / +45.0204	One liquid at x_1=0.02	0
0.12	Metastable / +3.4697	Two liquids	0.00821636
0.50	Unstable / -2.0000	Two liquids	0.11519417

For the two-liquid runs, x^{α}=**0.07072018** and x^{β}=**0.92927982**. They stay fixed when only z changes. At z=0.02, those coexistence endpoints are not a feasible split for that feed: the formal fβ≈-0.059076 is negative. Report one liquid, not two liquids with a negative amount.

G3. Independent fractions and balances

At z=0.12: fβ=(0.12-0.07072018)/(0.92927982-0.07072018)=**0.05739825**; fα=**0.94260175**. Balance: 0.94260175(0.07072018)+0.05739825(0.92927982)≈**0.12000000**. At z=0.50 both fractions are 0.50 and the balance is 0.50.

G4. The metastable feed can lower its energy

Using m(x)=x ln x+(1-x)ln(1-x)+3x(1-x), m(0.12)=**-0.05012499**. At both endpoints m≈**-0.05834135**, so the weighted split has this same energy. The decrease is **0.00821636** per RT. Positive curvature at 0.12 gives local resistance to infinitesimal changes, but a finite split lowers total Gibbs energy.

The supporting line describes realizable mixtures only between its endpoints. Extending it to z=0.02 violates nonnegative amounts. For actual coexistence, combine the balance, distinct endpoints, chemical-potential agreement and global tangent evidence. The Lab does not predict nucleation barriers or separation rates.

H-I. Transfer checks and reflection

Solutions page 4 | Do not read this page until you have attempted all H questions.

H1. One new residual

Central slope=[g(0.40)-g(0.20)]/(0.40-0.20)=(0.48-0.32)/0.20=**0.80**. Subtract L(0.30)=0.75 to obtain **r=+0.05**. This is a single approximate point residual, not the average across a dataset and not evidence about missing endpoint intervals.

H2. Missing isobaric information

Record **not assessed** for the heat-corrected local consistency test. Independent h^E is required for $h^E/(RT^2)dT/dx$. A raw score of 8.4 cannot be treated as an applicable failed local criterion when that required correction is unavailable. The number can still be shown as an uncorrected diagnostic.

H3. New endpoints and feed

fβ=(0.39-0.15)/(0.75-0.15)=**0.40**; fα=**0.60**. The balance is 0.60(0.15)+0.40(0.75)=**0.39**. Nonnegative amounts and a correct balance make the split feasible, not necessarily stable.

Confirm distinct phases, chemical-potential/common-tangent conditions and a tangent supporting the entire admitted Gibbs-energy curve. The candidate must be the global minimum subject to the balance at the specified T,P. Endpoints supplied without this evidence are only candidates.

H4. Three numerical results, no universal badge

Diagnostic	Given result	Interpretation under its assumptions
Isothermal area	D=1.2%	Meets D<5%
Local average	Score=8.4	Outside score<5
Selected model	Δp=0.3%; Δy=0.2 pp	Both reproduction criteria met

A close fit does not cancel a local discrepancy; the diagnostics ask different questions. State the disagreement and inspect its location and assumptions. Herington remains not applicable to the hypothetical isothermal path. None of these numbers alone identifies the cause as measurement error.

I. A useful anonymous reflection

In the Session 3 concept answer, explain the LLE evidence, especially the distinction between metastable homogeneous feed and stable two-liquid equilibrium. In the evidence answer, give a reproducible Lab 03 case and explain the area/local comparison, including the sampling limitation. During the slide 43 pause, complete feedback and include your actual study minutes and any pacing difficulty in the existing feedback text. Submit once and wait for **Response received**. On slide 44, verify that receipt and save your corrected work; do not resubmit. This form does not establish individual attendance.

Primary method descriptions: NIST area/Herington; NIST point test; NIST Van Ness. Lab 03 uses explicitly documented teaching variants. Its local operator is a three-point derivative of g, not NIST TDE's fitted Padé point test. Reference numerical results were checked against the shipped Lab engines.